

# Moshe meeting — what we have, what we don't

English version of `MEETING_MOSHE.md` . Same numbers, same order.

**One rule:** the numbers in this file are all of what has been measured and verified on the current backbone as of today. Do not quote a number that is not in here. In particular, `MEETING_PREP.md` , `MEETING_PREP_TR.md` , `CALL_NOTES.md` and `CALL_SCRIPT.md` quote **archived IQNet-era results that no longer hold** — nothing from those files.

## 0. The project in one sentence

"The benchmark number is not the contribution. Reproducing RadioML accuracy is solved. What I am measuring is *why* a classifier that reaches 99.9% on the training domain drops to 80% on IQ generated by a different transmitter — and I attribute it by intervention on the signal, not by correlation on the features."

*Note:* say this in the first 30 seconds. Everything else hangs off it.

## 1. WHAT WE HAVE — measured, with the file to show for it

### 1.1 The domain-gap table — the headline result

`compare_methods_ICRNNA_es.npz` · ICRNNA · 5 shared classes (BPSK, QPSK, 8PSK, 16QAM, 64QAM) · 5 seeds · 60-epoch ceiling / patience 20

| method                                    | in-domain | cross-domain        | gap           | 16QAM        |
|-------------------------------------------|-----------|---------------------|---------------|--------------|
| none                                      | 0.999     | <b>0.804</b> ±0.005 | <b>+0.195</b> | <b>0.060</b> |
| literature augmentation                   | 1.000     | 0.806 ±0.003        | +0.194        | 0.036        |
| <b>whitening <math>\alpha=0.75</math></b> | 1.000     | <b>0.993</b> ±0.001 | <b>+0.007</b> | <b>0.991</b> |
| whitening + augmentation                  | 0.999     | 0.994 ±0.003        | +0.005        | 0.998        |

"A model at 0.999 in-domain falls to 0.804 on an independently generated domain, and 16QAM — the class the whole diagnosis centres on — collapses to 0.060. The augmentation set from the literature, rotation and conjugate flip and additive noise, moves that gap by 0.001. Spectral whitening moves it from nineteen points to less than one, and costs nothing in-domain."

*Note:* the table is **complete at 20 of 20** and the file is in the repository. But the fourth row beats the third by **+0.001**, which is not a measurement. Adding the literature augmentation on top of whitening contributes nothing; whitening is doing all of the work. That is the opposite of what the file's own docstring predicted, and the prediction was written before the measurement, so it stays as a refuted hypothesis.

### 1.2 Plain classification — evidence the backbone is competent

| run                                   | overall               | SNR $\geq$ 10 dB      | status                    |
|---------------------------------------|-----------------------|-----------------------|---------------------------|
| RML2016.10a, 11 classes, 3 seeds      | <b>0.6223</b> ±0.0019 | <b>0.9148</b> ±0.0013 | converged (best epoch 39) |
| RadioML 2018.01A, 24 classes, 3 seeds | <b>0.5699</b> ±0.0081 | <b>0.8716</b> ±0.0141 | converged (best epoch 26) |

For 2018, 512 of the 4096 frames per (class, SNR) cell, at the native 1024-sample frame length.

"These are not the contribution — they exist so that the cross-domain numbers are known to start from a competent classifier rather than a broken one."

*Note:* almost all of the 2016 error is **one cell**. WBFM is read as AM-DSB 60% of the time, and that single cell is 64% of every misclassified frame at SNR  $\geq 10$  dB. Drop WBFM and the other ten classes sit at 0.968. This is a property of the dataset, not of the model — both classes are generated from voice recordings, and during the silences in the source audio both degrade to an unmodulated carrier.

### 1.3 Against the published paper — the reproduction question is closed

`colab/icrnna_faithful_2016.py`, built from the paper (El-Haryqy et al., *Results in Engineering* 26 (2025) 104783) rather than from a reproduction:

|               | 58-epoch ceiling (the paper's own budget) | 150-epoch ceiling                   |
|---------------|-------------------------------------------|-------------------------------------|
| best epoch    | 58, 58, 55 — <b>at the ceiling</b>        | <b>107</b> — early stopping decided |
| test accuracy | 61.75% $\pm 0.07$                         | <b>63.21%</b>                       |
| paper: 63.24% | −1.49 points                              | <b>−0.03 points</b>                 |
| parameters    | 794,827 (paper: 0.79M) ✓                  | same                                |

"The faithful build reproduces the paper — 63.21 against 63.24. It needed nearly twice the published epoch count to get there, and that was the whole of the deficit. The paper's 'Number of Epochs: 58' is a report of where its own early stopping landed, not a recipe that transfers."

### 1.4 A methodological finding — worth raising

**Convergence is set by gradient updates, not epochs.** 2016 converged in 23,439 updates and 2018 in 22,672 — within 3% of each other, despite differing in class count and in sequence length by a factor of eight.

But that is not a constant. Inside `compare_methods` the stopping epoch depends strongly on the **method**:

| method                  | stopping epochs    | mean | updates       |
|-------------------------|--------------------|------|---------------|
| none                    | 51, 41, 48, 36, 38 | 43   | 9.8k – 13.9k  |
| standard augmentation   | 89, 48, 83, 65, 82 | 73   | 13.1k – 24.3k |
| whitening $\alpha=0.75$ | 26, 26, 33         | 28   | 7.1k – 9.0k   |

"Whitening converges 2.6× faster than the augmentation row. That is what the mechanism predicts — whitening removes a nuisance dimension so there is less to fit; augmentation adds one so there is more. It also means a single epoch ceiling cannot be assumed safe across a sweep whose cells differ in method."

### 1.5 Causal attribution — four refuted hypotheses, one working intervention

The gap concentrates almost entirely in one cell: 16QAM read as 64QAM. Four explanations were **refuted** by controlled experiment:

| hypothesis                                         | outcome                                                   |
|----------------------------------------------------|-----------------------------------------------------------|
| occupied bandwidth differs                         | refuted — matching bandwidth did not recover 16QAM        |
| constellation density                              | refuted — density is not what the model reads             |
| symbol-timing regularity                           | refuted — RadioML has <b>no</b> symbol-rate spectral line |
| channel impairments (CFO, multipath, IQ imbalance) | refuted — removing them one at a time changed little      |

What worked: **phase-preserving magnitude substitution** ( `narrow_the_search.py` ). Take a frame from the failing domain, keep its phase spectrum exactly, and substitute only the magnitude spectrum of the training domain. The prediction follows the magnitude spectrum.

"That is an intervention on the signal, not an observation of the representation — so it supports a causal claim rather than a correlational one. The cause is the transmitter's spectral envelope, chiefly its pulse-shaping roll-off."

*Note:* these are signal-level and independent of the classifier, so they still stand. But the **accuracy numbers attached to them are pending re-measurement**.

## 1.6 Other valid results that involve no model

- `baseline_results.npz` — the classical cumulant + SVM pipeline, 11 modulations, peak accuracy 0.974. No neural network, still valid.
- `sink_vs_geometry.npz` — signal geometry, no model, still valid.
- `overfit_2x2.json` — architecture comparison (IQNet vs ICRNNA × fixed vs plateau LR), still valid.
- The **signal-level** figures in `figures/` (spectrograms, constellations, class spectra) are valid. Anything showing an accuracy curve or a confusion matrix **predates the current backbone — do not use it**.

## 2. WHAT IS RUNNING NOW

- `compare_methods_ICRNNA_es_e100.npz` — the same table at a 100-epoch ceiling. It stands at **15 of 20 cells**; the whitening + augmentation row is what is left. Three finished cells still stopped because the budget ran out (peaks at 89, 83, 82, which needed 109, 103, 102), all of them in the augmentation row. Those get `--epochs 150 --redo-unconverged`.
- It does **not** reproduce the 60-epoch table cell for cell — that one was measured on the laptop and this one on Colab. They are independent samples, so a cell-level difference between them is hardware, not the ceiling.
- The cost is small: **the ceiling only matters to a cell that hits it**. A cell that peaked at epoch 36 peaks at 36 whatever ceiling it ran under. Ten cells rerun, not twenty.

### 3. WHAT WE DON'T HAVE — say it before being asked

| missing                                                                     | why it matters                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>the <math>\alpha</math> sweep</b><br>( <code>whitening_seeds.py</code> ) | $\alpha=0.75$ is <b>unverified</b> for the current backbone. The claim that partial whitening beats full whitening — our one concrete disagreement with WhiteNet — currently rests on the old measurement. 25 cells, not run. |
| <b>the sink / family thread (24-class leave-one-out)</b>                    | The <b>most novel</b> part of the work: an unseen modulation lands inside its own family, checked five independent ways. <b>Never rerun</b> on the current backbone. The most expensive job.                                  |
| <code>dann.py</code>                                                        | Built on the legacy feature extractor, not ported. Its numbers are comparable to nothing. Either rewrite it or drop the comparison.                                                                                           |
| <b>real SDR capture</b>                                                     | <b>The single largest weakness.</b> Both domains are synthetic. This is why the work is a study rather than a result. Needs hardware and lab access.                                                                          |

*Note — do not defend the fourth row, concede it directly:*

"The single largest weakness of this work is that both domains are synthetic. RadioML is simulated and so is my generator. Until I can do an over-the-air capture this is a study of a mechanism, not a measurement of a deployment. That is also the thing I would most like to fix, and it is a hardware and lab access question rather than a compute question."

### 4. CAVEATS TO STATE UP FRONT

Do not make him find these. Say them. The credibility comes from here.

1. **The current model is not the published architecture.** `model_zoo.ICRNNA` was transcribed from a peer's reproduction; against the paper it differs in **five places** (conv1 kernel 5 vs 3, two max-pools vs one, one BatchNorm after the LSTM stack vs one per layer, no attention dropout or LayerNorm, one dense layer of 128 at dropout 0.5 vs two of 128 and 64 at 0.3). Do not call it "the published architecture". The faithful build is a separate file and **does** reproduce the paper (§1.3).
2. **Whitening is not novel.** WhiteNet (arXiv:2608.06581) arrived at essentially the same operation, and on **real** over-the-air captures. The novelty claim was dropped on finding it. What survives: the attribution method, the partial-vs-full disagreement, and the sink finding. It is recorded in `LITERATURE.md`.
3. **The mechanism is not fully resolved.** The working account through most of the project was shortcut learning: the model reads modulation order off the spectral envelope. **Two experiments killed it** — most directly the envelope transplant, where handing 16QAM frames a 64QAM envelope left the prediction at 16QAM. The envelope does not *carry* the class; removing it still helps. That contradiction is stated **as unresolved**.

"I would rather tell you the part I cannot explain than have you find it."

4. **The small differences in the table are not measurements.** At a 60-epoch ceiling at least two cells stopped without converging. The 0.804 → 0.993 headline is unaffected (19 points against a seed spread of 0.005). But differences the size of whitening vs whitening+augmentation (+0.001) or none vs augmentation (+0.002) are **not reportable** until the e100 table is finished.
5. **Three seeds is thin on 2018.** Seed spread reaches 0.021 on 24 classes. A difference of less than about 0.03 between two configurations measured this way **is not yet a difference**.

---

## 5. WHERE WE WANT HELP — what to ask Moshe

If anything is to come out of the meeting, it is these:

1. **SDR access.** "Both domains are synthetic" is the largest single-sentence hole. A USRP or HackRF, lab access, and guidance on the capture protocol — this is the one thing that moves the work from a study to a result.
2. **The open-set recognition literature.** That is the field adjacent to the sink finding, and a few searches do not cover it. The novelty question there is genuinely open. Who should I be reading?
3. **Which piece first?** Compute is limited. The  $\alpha$  sweep (cheap, confirms the WhiteNet disagreement) or the 24-class leave-one-out (expensive, most novel)? An outside opinion is worth having here.
4. **Target venue.** The contribution is an attribution method, not a benchmark number. Does that belong in a signal processing venue or a machine learning one?

---

## 6. AFTER THE MEETING — immediate

- ☒ Pulled the completed `compare_methods_ICRNNA_es.npz` (20/20) from Drive into the repository, over the old 17/20 copy; every shared cell was bit-identical.
- ☐ When `compare_methods_ICRNNA_es_e100.npz` finishes, rerun the three augmentation cells with `--epochs 150 --redo-unconverged`.
- ☐ The  $\alpha$  sweep ( `whitening_seeds.py` ) — seed count to be decided against the compute budget.
- ☐ Either label the old meeting files ( `MEETING_PREP*.md` , `CALL_NOTES.md` , `CALL_SCRIPT.md` ) clearly as IQNet-era or move them to the archive. As they stand they are a trap.